

Predicting Movie Popularity: A Self-Collected Dataset for Hit/Flop Classification

1. Title and Collection Method

This dataset, the *Movie Popularity Prediction Dataset*, was built to explore whether characteristics of a movie that are known before or shortly after release can help anticipate how an audience receives it. Data was collected through a Google Form distributed to a group of student contributors.

2. Description of Features and Label

Input features (X):

- **Ratings (1–10):** audience rating score for the movie.
- **Duration (minutes):** the movie's runtime.
- **Genre:** the primary genre category (e.g. Action, Horror, Thriller, Drama).
- **Country:** the country (or countries) of production.
- **Year:** the release year of the movie.

A sixth column, **Number of Votes**, was also collected but is deliberately excluded from X. It is used only to derive the label below; including it as an input feature would leak the label into the data and make the prediction problem trivial.

Label (y): Popularity (Hit/Flop). This is a binary classification target, computed automatically rather than guessed: each movie's vote count is compared to the **median** vote count across the full sample. Movies at or above the median are labeled *Hit*; movies below it are labeled *Flop*. For the current 23-row sample, the median — and therefore the Hit/Flop threshold — is 240,000 votes. This threshold is relative to the sample and will shift automatically as more rows are added.

3. Dataset Structure

The dataset currently contains **23 rows** (one row per movie) and **8 columns** (Movie Name, Ratings, Duration, Genre, Country, Year, Number of Votes, Popularity). The table below shows the first 10 rows exactly as collected, including their natural messiness (blank cells, a placeholder “?”, and a duplicate movie entry) — nothing has been cleaned or altered.

Movie	Rating	Dur.	Genre	Country	Year	Votes	Pop.
John Wick	10	154	Action	USA	2014	2,400,000	Hit
Parasite	9	159	Thriller	South Korea		240,000	Hit
Train to Busan	8	124	Horror	South Korea	2016	2,400	Flop
Get Out	8	180	Horror	USA	2017	120,000	Flop
RRR		140	Action	India		2,300,000	Hit
The Dark Knight	7	160	Action	USA	2008	240,000	Hit
Silence of the Lambs	6	167	Thriller	USA	1991	134,324	Flop
Oldboy	5	154	Thriller	South Korea	2003	2,800,000	Hit
The Matrix	10	130		?	1999	342,903	Hit
RRR	9	170	Action	India	2022	1,820,000	Hit

4. Quality Issues

Reviewing all 23 rows (not just the sample above) surfaced several concrete data quality issues, typical of multi-contributor manual data collection:

- **Duplicate entries:** “John Wick” appears 3 times and “RRR” appears 2 times, each time with different rating, duration, and vote-count values — the result of different contributors independently researching the same title from the shared dropdown list rather than coordinating who covers what.
- **Missing values:** Ratings are missing in 2 rows, Duration in 4 rows, Genre in 2 rows, Country in 2 rows, and Year in 7 rows (5 true blanks plus 2 cases described below).
- **Placeholder text instead of true blanks:** two rows contain a literal “?” in the Genre/Country fields, and two rows contain the literal text “NaN” (typed as a word, not a true empty cell) in the Year/Duration fields. These must be standardized to genuine missing values before any statistics are computed on those columns, since spreadsheet and analysis tools will otherwise treat “?” or “NaN” as a real category or string rather than recognizing it as absent data.
- **Inconsistent formatting:** the Country field contains a leading-space variant (“ South Korea”) alongside the correctly formatted “South Korea” elsewhere in the same column — a whitespace inconsistency that would cause the two to be treated as different categories unless trimmed.
- **Suspect / erroneous entry:** one row records the movie name as “Option 15” rather than an actual title, most likely caused by a mismatch between the Google Form's internal option values and its displayed labels after the choice list was edited mid-collection.
- **Small sample size:** at 23 rows, the dataset is below the 50-row target for this assignment. Continued collection is needed before this dataset should be treated as final.

5. Learning Type

This is a **supervised learning** problem. Every row has a clearly defined target outcome — Popularity (Hit/Flop) — and that label was derived independently of the input features (from vote count relative to the sample median, not from the rating or any X variable). Because a known, correct answer (y) exists for each training example, a model can be trained to map the input features X (Rating, Duration, Genre, Country, Year) to this known output, which is the defining characteristic of supervised learning. Specifically, since the label is categorical with two classes, this is a **binary classification** problem rather than regression.

It is worth noting that the same five input features, with the label removed, could alternatively support an **unsupervised** task — for example, clustering movies into natural groupings (by genre, country, and era) without reference to popularity at all. However, since this dataset does include a clear, independently-derived y, the supervised framing is the appropriate one here.

6. Use Case and Data Science Lifecycle

In a machine learning project, this dataset would support a **binary classification** task: predicting whether a movie is likely to be a Hit or a Flop based on attributes available around release time (rating, duration, genre, country, year). A streaming platform or studio could use a model like this to flag titles worth additional marketing investment, or to benchmark a new release against historically similar films.

Within the data science lifecycle, this dataset and paper represent the **data collection and data understanding** stage: defining the problem, identifying what raw data is needed, gathering it from real sources, and documenting its structure and flaws. The next stage — **data preparation / preprocessing** (Lesson 3) — will involve resolving the duplicate rows, converting placeholder text (“?”, “NaN”) and blanks into proper missing-value handling, trimming whitespace inconsistencies, encoding the categorical Genre and Country columns, and scaling the numeric features, before the dataset is ready for model training and evaluation.